

FACILITATION PAPER 2025 PHYSICS

SECTION A

ITEM 1

Your area Member of Parliament has organized a camp in a nearby playing ground that is located between a tall building and a cliff to give feed back to his people. He is also required to donate some equipment's to boda boda riders who drive during the day and night in a bid to reduce accidents they normally meets on the road while on their motorcycles. Among the items to select from by boda boda riders were concave mirrors, convex lenses, small plane mirrors.

During the presentations, different reactions and observations were made and noticed by the Member of Parliament.

- i. One herds man threw a grazing stick to a cow that accidentally fell in a nearby a pond with clear waters. As he proceeded to pick it he found out that a big part of it appeared as if it's broken and his Appearance appeared in water. This confused him.
- ii. One boda boda rider required for some mirrors to use while riding and the front lighting system as the ones originally on the moto cycle were spoilt when it fell down.
- iii. Some people in attendance are worried that the sound produced by the loud speakers could pose risks to their ears because its too loud, un clear and disorganized.

Fig 1 (a) shows the rarefactions and compressions produced by the sound wave heard

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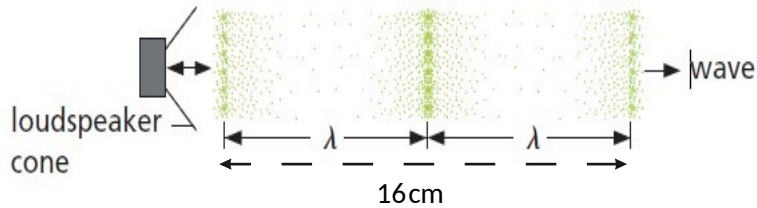


Figure 1(b) shows the wave signal that produces the sound

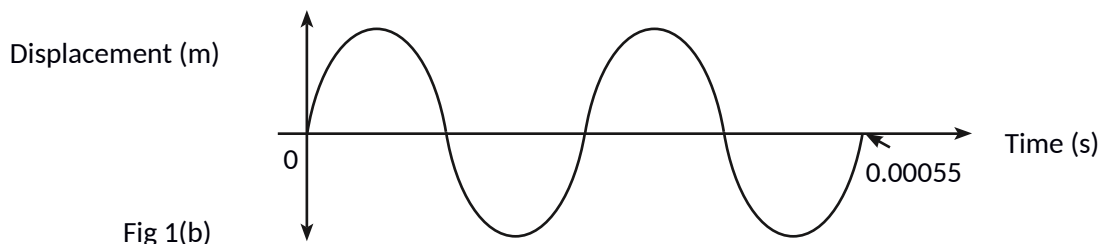


Fig 1(b)

Audible frequency range = 20 - 20000 Hz

Use speed of sound in air = 330 ms^{-1}

Task

- (a) To arrest the herds man concerns/ fears
- (b) Select with reason, the most appropriate mirror equipment to donate to the boda boda rider.
- (c) Understand whether the sound produced posed risks to the attendants' ears

(d) why the sound seemed unclear and disorganized.

(e) Describe one area in which one of the concepts can be applied in daily life.

Item 2

A school staff had visited the coastal town of Indian Ocean during months of May to October. They ring their driver to come and pick them as they wish to return back to Uganda, but the driver does not know well their exact location. He asked from them to get direction of where to find them. Immediately they send him coordinates using an application on the phone. Upon arrival, the driver develop an interest to swim but was cautioned that it's risky during that time. The driver was left wondering why it's not safe/ risky.

Task:

Help to explain to the driver;

(a) Why he was stopped to swim at that time.

(b) How he was able to locate the staff using the coordinates sent to him.

(c) Describe one area in which one of the concepts in one area of the challenges can be applied in daily life.

Item 3

In a certain town, the amount of water reaching people for use in homes has reduced and they are worried of water scarcity in the near future if this is not rectified.

The authorities would like to rule pipe but don't know it's the exact spoilt or suspects a leakage in underground.

They have been told that radioactive material should be used to check the spot but they have little knowledge of doing it.

Table 1 shows the materials available for use in the process of checking if there is leakage underground.

Detector	Counter
Spark counter, cloud chamber geiger muller tube	Rate meter, scaler.

A sample of Technetium-99m of activity 200 counts per minute is used in the detector until its activity reduces to background radiation of level of 50 counts per minute.

The locals have been asked to wait for sometime when this material is used and also to take precautionary measures of dangers associated with the use of the material but are left wondering when.

Hint

Half life of Technetium is 6 hours

Task.

You have been approached by both locals and local authorities, help them with relevant information regarding to the challenges faced.

SECTION B

Attempt one item only.

ITEM 4

In a certain factory, different loads are transferred from one point to another using a conveyer or belt.

The length of the conveyor is 14cm and is driven by a constant force of 15N. the system is supplied by cold water from the drum connected to it by a series of pipes for efficient running of the machine but was observed that the level of water kept reducing and this worried operator. During the operation, an unusual noise was heard from access of the machine and some parts felt hot when touched. The operator is worried the origin of the noise and heat and how it can be overcome.

He is told that the system is able to operate at 50% efficient but is wondering how much load can be carried by conveyor without the challenges identified.

Task.

(a) Help the operator understand:

- (i) The cause of the heat and noise observed during operation and how they can be minimized.
- (ii) The maximum load that can be carried by the conveyor without the challenges identified.

(b) Why the water supply tank abruptly stopped working and what can be done to rectify the problem.

- (c) State one area in which the one the concepts in one area of the challenges can be applied in daily today lives.

ITEM 5

While at home with your brother painting the house, Janie came to visit you and accidentally leaned on the wall which had just been painted which stained her uniform. Janie was given different advices to remove the stains as:

1. To put paraffin/ kerosene on stained part and it was observed that the paraffin spread immediately on the cloth but could not remove it.
2. To wash the stained part with water and soap/omo but could not also remove it.

Your brother finally advised her to use warm water at a temperature of $46-51^{\circ}\text{C}$ that this would easily remove the stain. This advise left both Janie and her friend wondering how this would be possible.

Hint

- A long capillary tube of about theft are for available about 28 cm with a bulb.
- A 12cm column of mercury may be appropriate to determine the temperature of water to be used.

Task

- (a) Help Janie understand;
- (i) Why paraffin spread on the whole cloth, and stains got off quickly when done as she was advised by your brother.

- (ii) Establish if it would be possible with warm water to remove the stains at the stated temperature.
- (iii) Explain the application of any one concepts to addressing the challenges in daily life.

PART II

Attempt one item.

Item 6

A group of students in a certain boarding school decided to revise for exams at night from a field north of the school. They were provided with a voltmeter three DC electric lamps resistance 12Ω each, an accumulator rating 12v and 8A, switch connecting wires, 3-inch steel nail, a bare copper wire of length 50cm and a piece of thread. However, the rest of your colleagues are unable to connect the circuit correctly except you. They also want to know the best ways of using the accumulator for a longer time. Further after revision, they could not easily tell the direction from which they came.

Task.

- (a) Advise on how to generate light so that you and your colleagues can revise for a longer time and state assumptions made.
- (b) Come up with advice that would enable you and your colleagues back to school using available materials.
- (c) State the applications of the concepts observed in the scenario.

Item 7

A factory that specializes in manufacturing metallic products uses a system/device to sort and separate different metallic raw materials. The device draws a current of 10 A during operations, efficiently segregating the materials to streamline production from the surface. Each system unit consumes 100 W of electricity per hour and typically operates for 6 hours a day. The manager wanted to understand the principle of operation of the system and why it is overheating and also noticed that heavy metals were not segregated. He has been advised to improve energy efficiency by decreasing energy consumption by 25%, and strengthen the device/system but he lacks the knowledge of doing, and whether there is any other application of the device

Task

As a student of Physics, assist the factory manager:

- (a) Understand the system and how it operates
- (b) Find out the energy required to run the system efficiently without overheating.
- (c) Explore how to enhance the system and any other application of it.

END